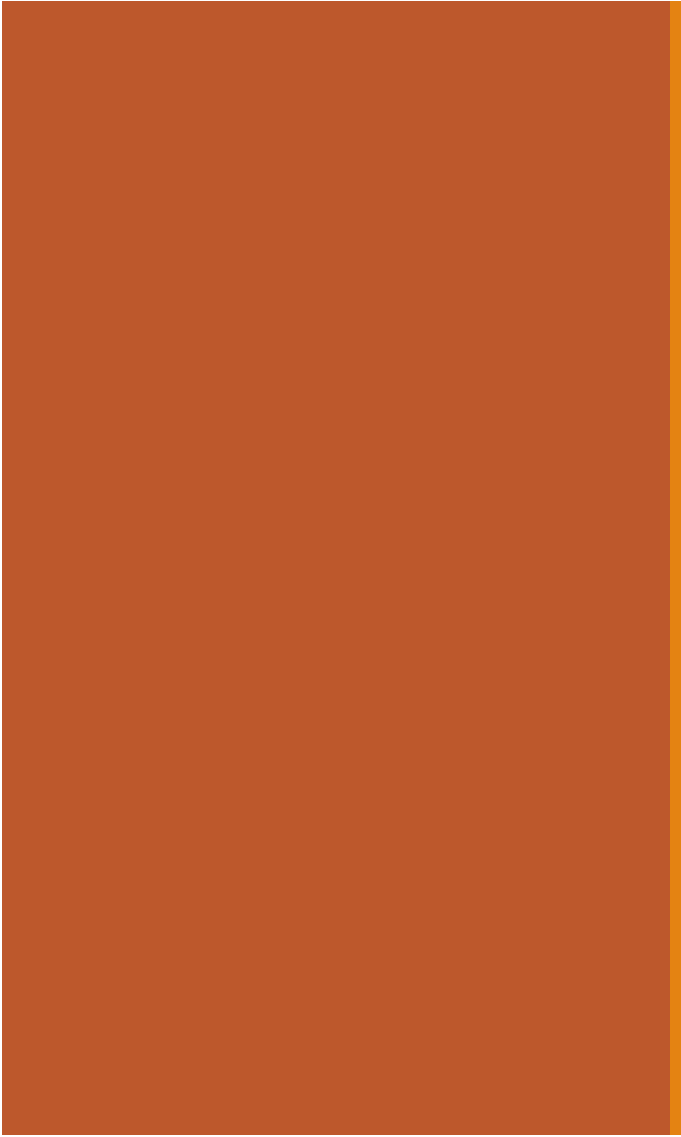


15: General Inference

Jerry Cain
May 3rd, 2024

[Lecture Discussion on Ed](#)



Inference

Inference

*Web*MD[®]

Inference

WebMD Symptom Checker WITH BODY MAP

INFO SYMPTOMS QUESTIONS CONDITIONS DETAILS TREATMENT

What are your symptoms?

Type your main symptom here

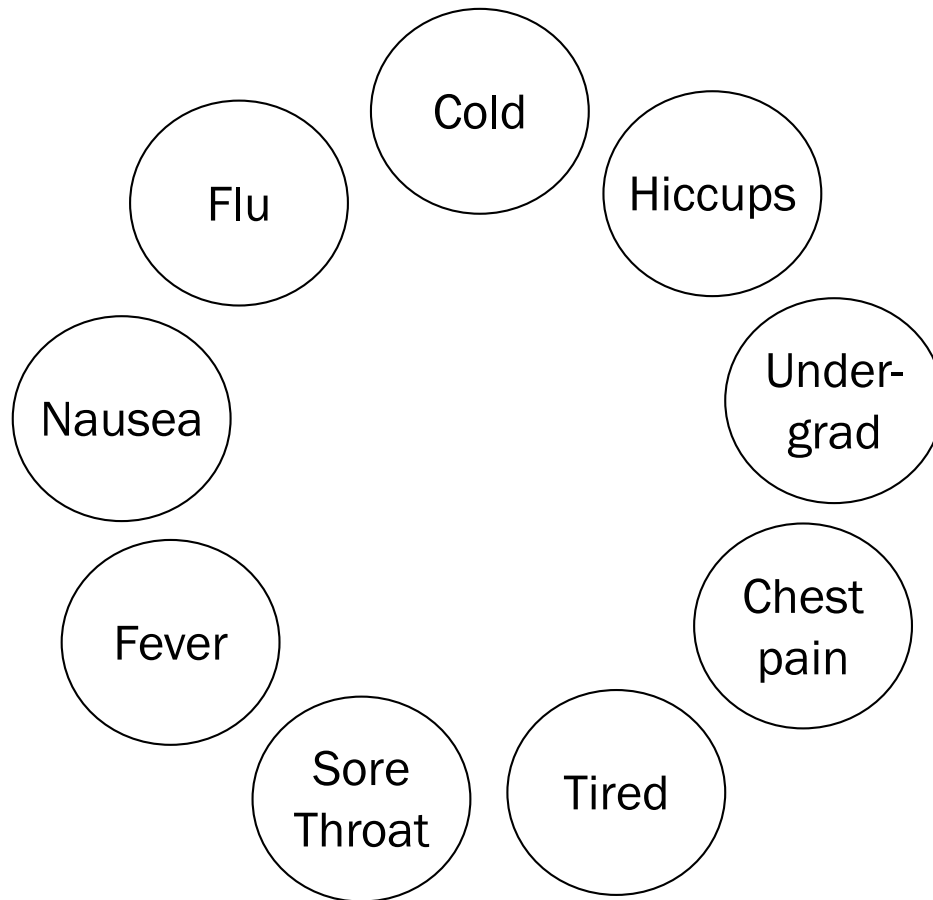
My Symptoms

- nausea
- fever 100.5f to 102f
- severe headache
- shaking chills

Results Strength: MODERATE



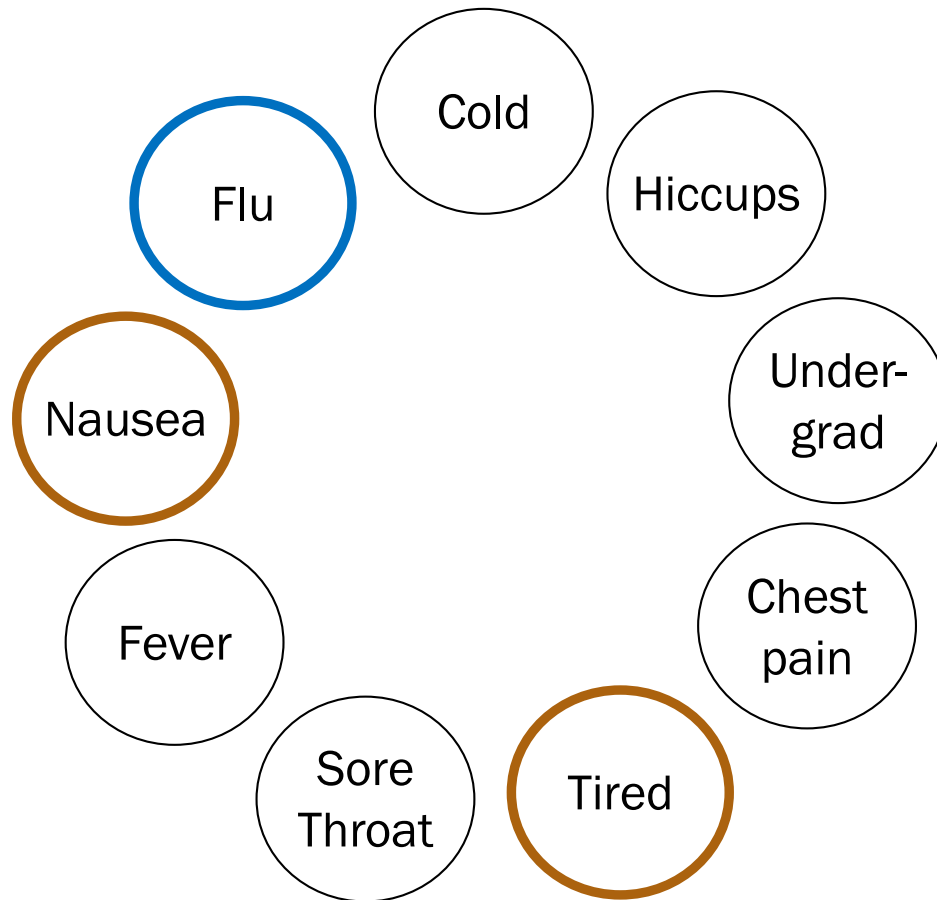
Inference



General **inference** question:

Given the values of some random variables, what is the conditional distribution of some other random variables?

Inference

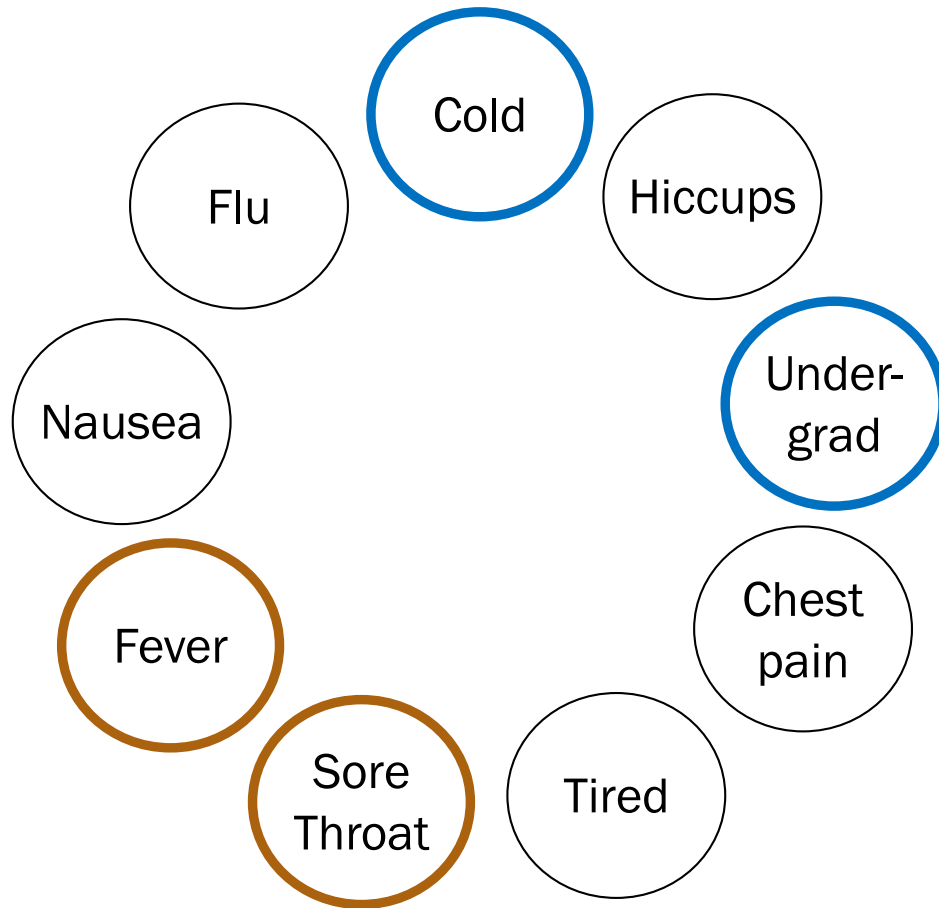


One inference question:

$$P(F = 1 | N = 1, T = 1)$$

$$= \frac{P(F = 1, N = 1, T = 1)}{P(N = 1, T = 1)}$$

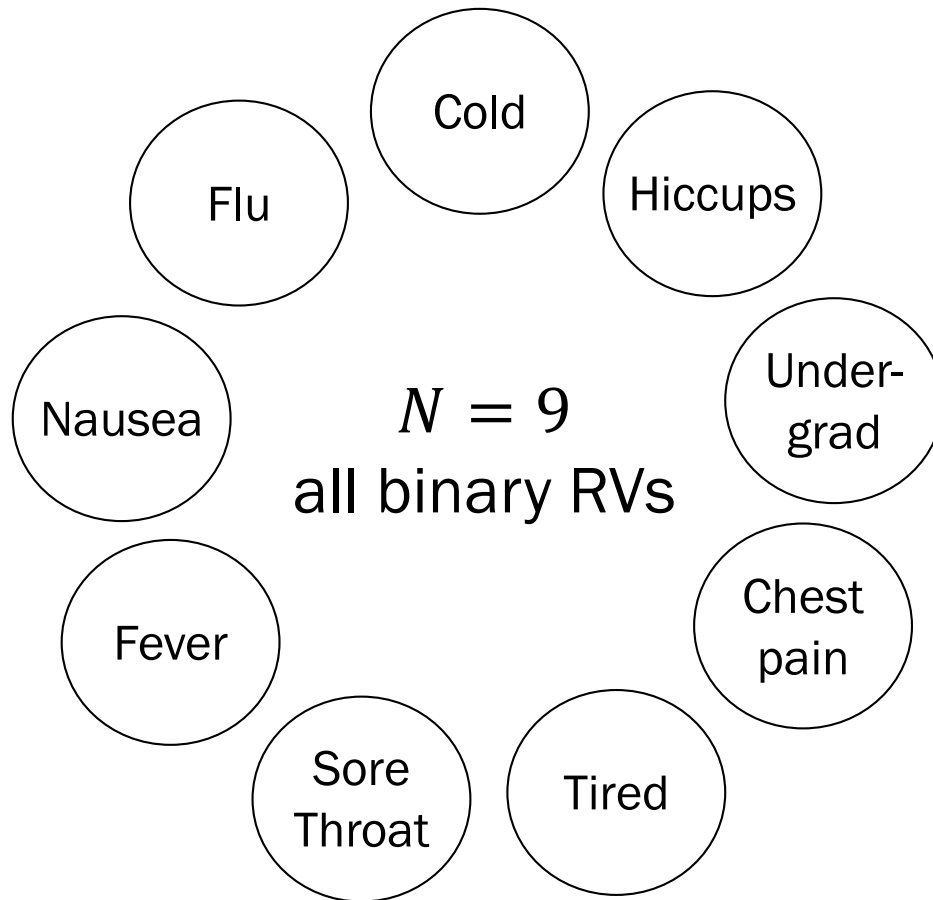
Inference



Another inference question:

$$P(C_o = 1, U = 1 | S = 0, F_e = 0) \\ = \frac{P(C_o = 1, U = 1, S = 0, F_e = 0)}{P(S = 0, F_e = 0)}$$

Inference



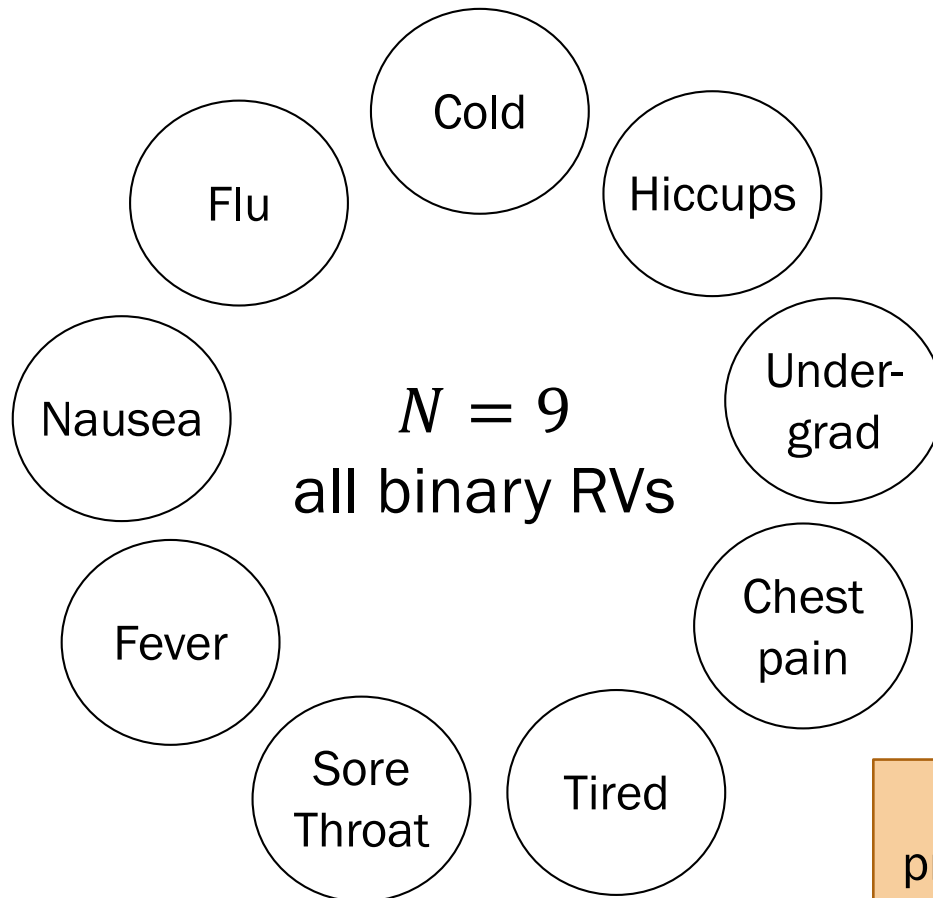
If we know the full **joint distribution**, we can answer all probabilistic inference questions.

What is the size of the joint probability table?

- A. 2^{N-1} entries
- B. N^N entries
- C. 2^N entries
- D. None/other/don't know



Inference



If we know the full **joint distribution**, we can answer all probabilistic inference questions.

What is the size of the joint probability table? *why 2^N ?*

A. 2^{N-1} entries

B. N^N entries

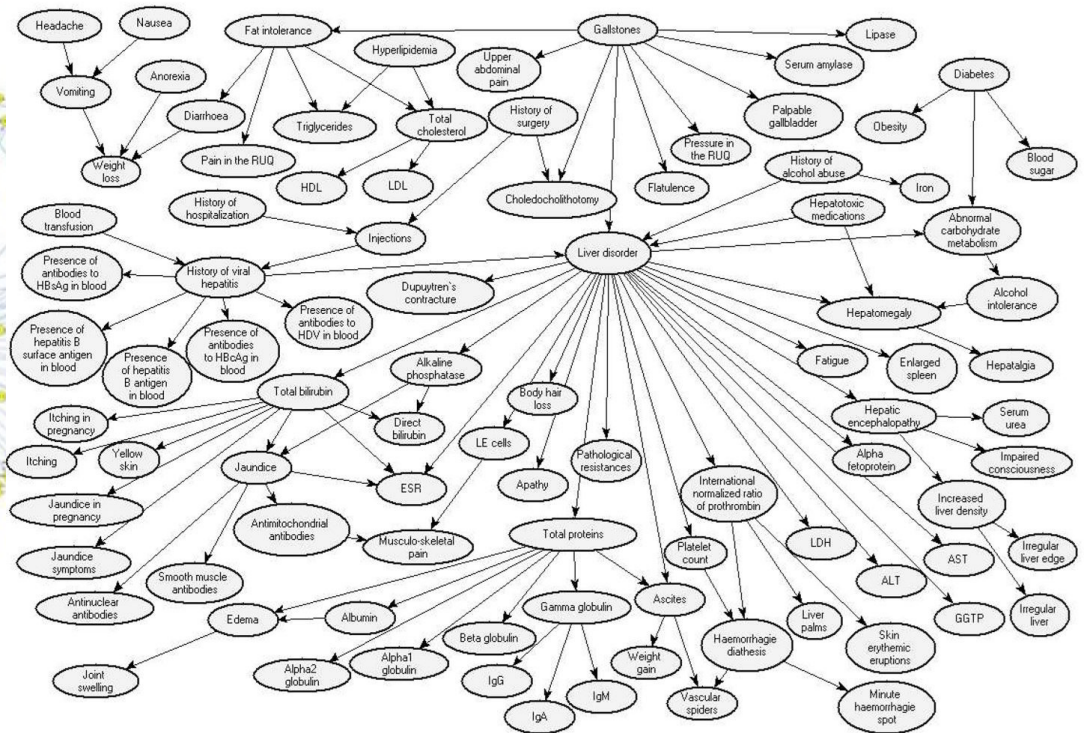
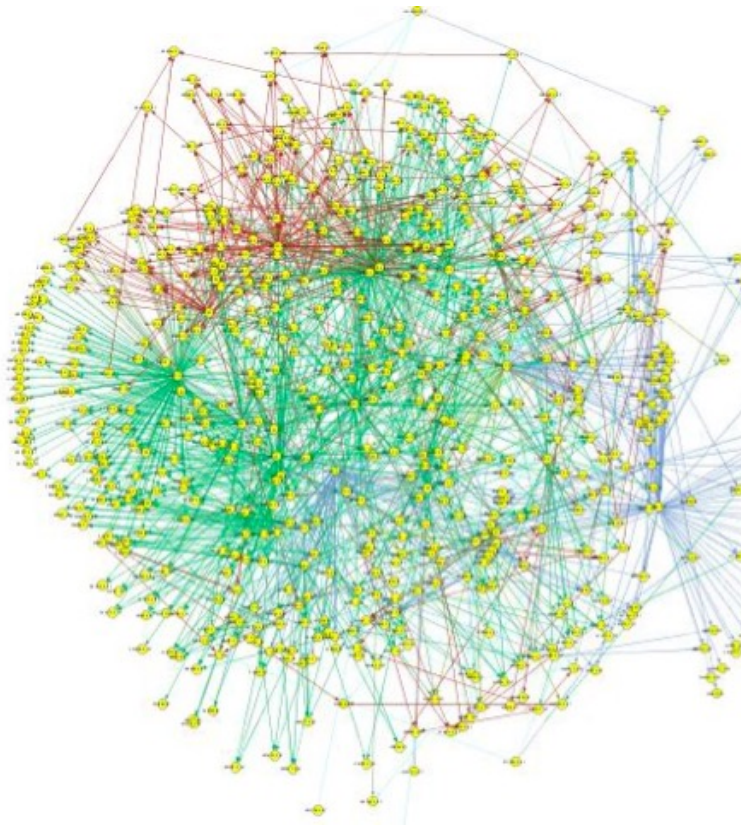
C. 2^N entries

D. None/other/don't know

$P(X_1 = _, X_2 = _, \dots, X_N = _)$
= some value that needs to be stored somewhere.
Each $_$ can be filled in with two values independently of the others.

Brute-force computation of a full joint probability mass function is often intractable.

100



Conditionally Independent RVs

Recall that two events A and B are conditionally independent given E if:

$$P(AB|E) = P(A|E)P(B|E)$$

n discrete random variables $X_1, X_2, \dots, X_n$ are called **conditionally independent given Y** if:

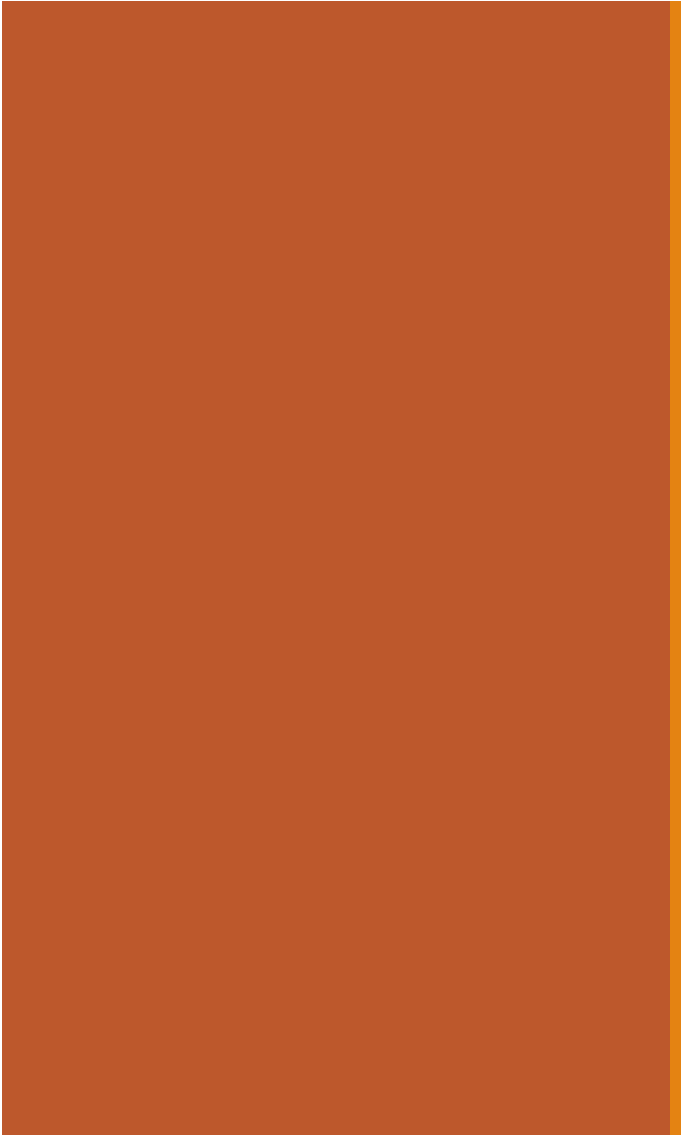
for all $x_1, x_2, \dots, x_n, y$: $\left. \begin{array}{l} \text{i.e. for any and} \\ \text{all combinations} \\ \text{of values.} \end{array} \right\}$

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n | Y = y) = \prod_{i=1}^n P(X_i = x_i | Y = y)$$

This implies the following (cool to remember for later):

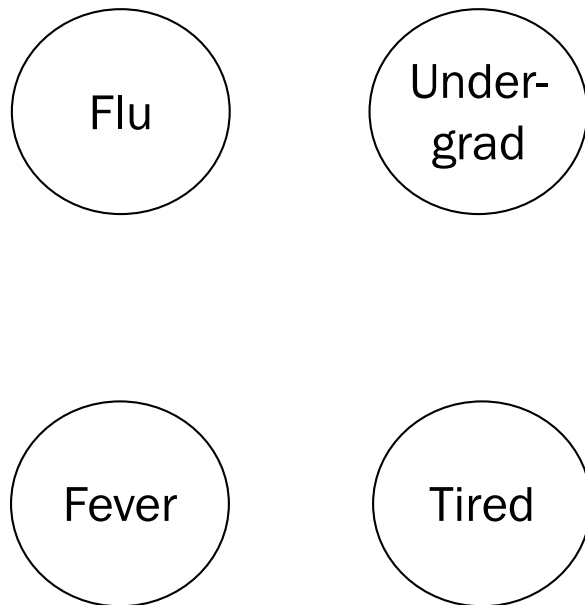
$$\log P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n | Y = y) = \sum_{i=1}^n \log P(X_i = x_i | Y = y)$$

this will be useful later on when probabilities are so small that multiplying them makes it difficult to maintain precision



Bayesian Networks

A simpler WebMD



Great! Just specify $2^4 = 16$ joint probabilities?

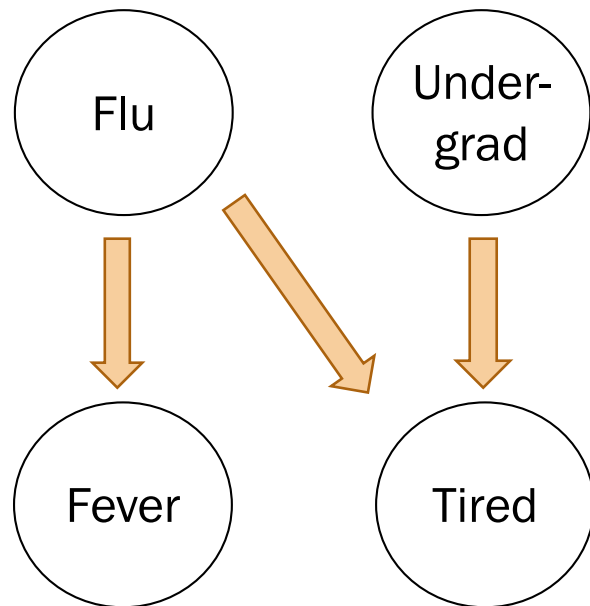
we're still assuming all random variables are binary / Bernoulli;

$$P(F_{lu} = a, F_{ev} = b, U = c, T = d)$$

What would an infectious diseases (ID) expert do?

Describe the joint distribution using causality!

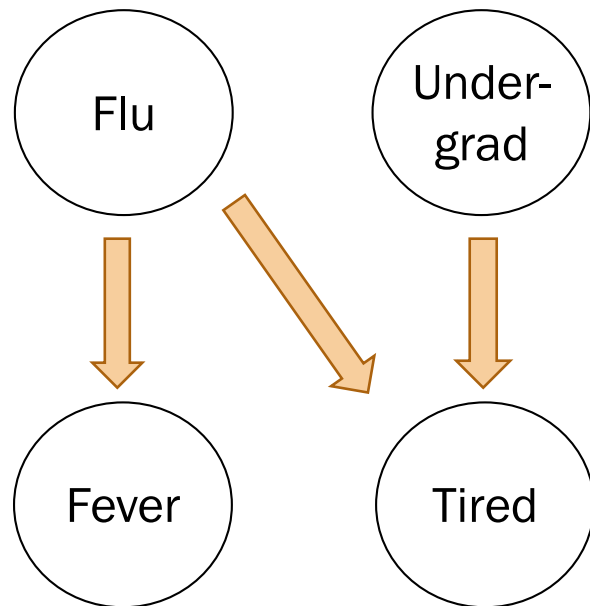
Constructing a Bayesian Network



What would an ID expert do?

1. Describe the joint distribution using causality.
2. **Assume conditional independence.**

Constructing a Bayesian Network



In a Bayesian Network,
Each random variable is **conditionally independent** of its non-descendants, **given its parents**.

- Node: random variable
- Directed edge: conditional dependency

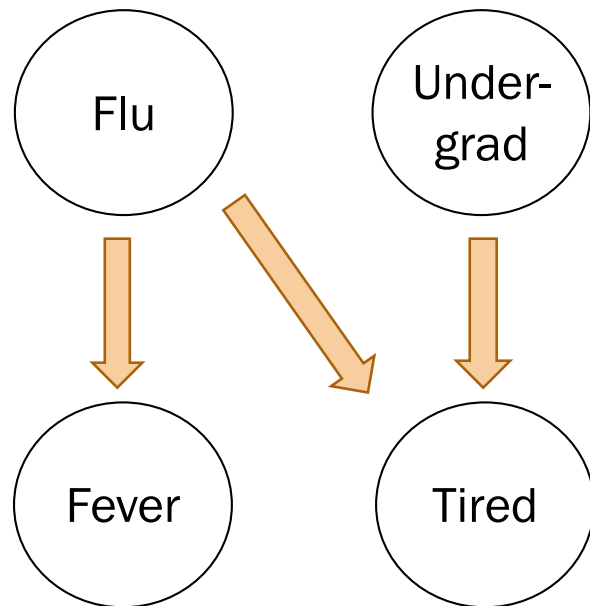
Examples:

- $P(F_{ev} = 1 | T = 0, F_{lu} = 1) = P(F_{ev} = 1 | F_{lu} = 1)$ *no influence on Fev*
- $P(F_{lu} = 1, U = 0) = P(F_{lu} = 1)P(U = 0)$ *no parents, so assumed to be independent*

Constructing a Bayesian Network

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

What would an ID expert do?

1. Describe the joint distribution using causality.
- ✓ 2. Assume conditional independence.
3. Provide $P(\text{values}|\text{parents})$ for each random variable

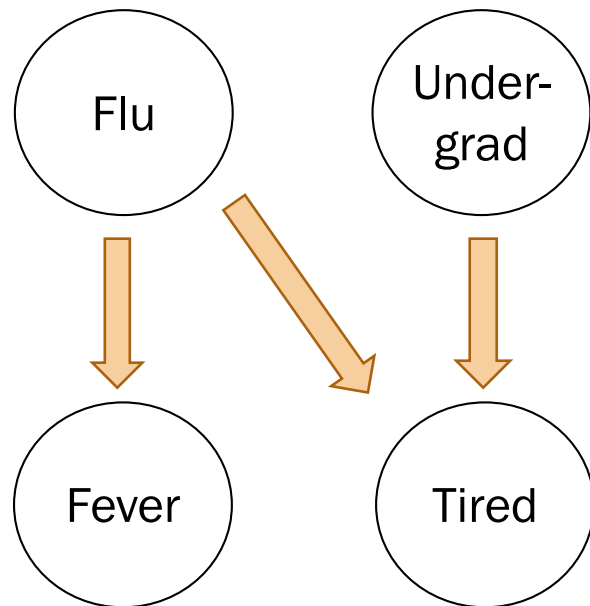
What conditional probabilities should our expert specify?



Constructing a Bayesian Network

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1|F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1|F_{lu} = 0) = 0.05$$

What would an ID expert do?

1. Describe the joint distribution using causality.
2. Assume conditional independence.
3. Provide $P(\text{values}|\text{parents})$ for each random variable

What conditional probabilities should our expert specify?

$$P(T = 1|F_{lu} = 0, U = 0)$$

$$P(T = 1|F_{lu} = 0, U = 1)$$

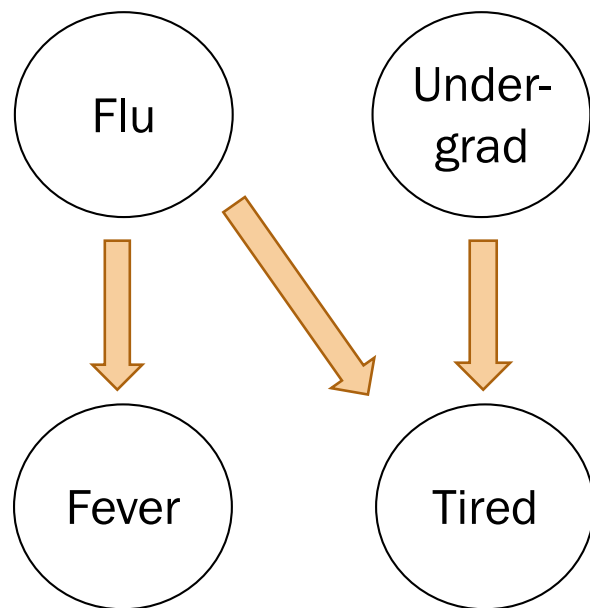
$$P(T = 1|F_{lu} = 1, U = 0)$$

$$P(T = 1|F_{lu} = 1, U = 1)$$

Using a Bayes Net

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1|F_{lu} = 1) = 0.9$$
$$P(F_{ev} = 1|F_{lu} = 0) = 0.05$$

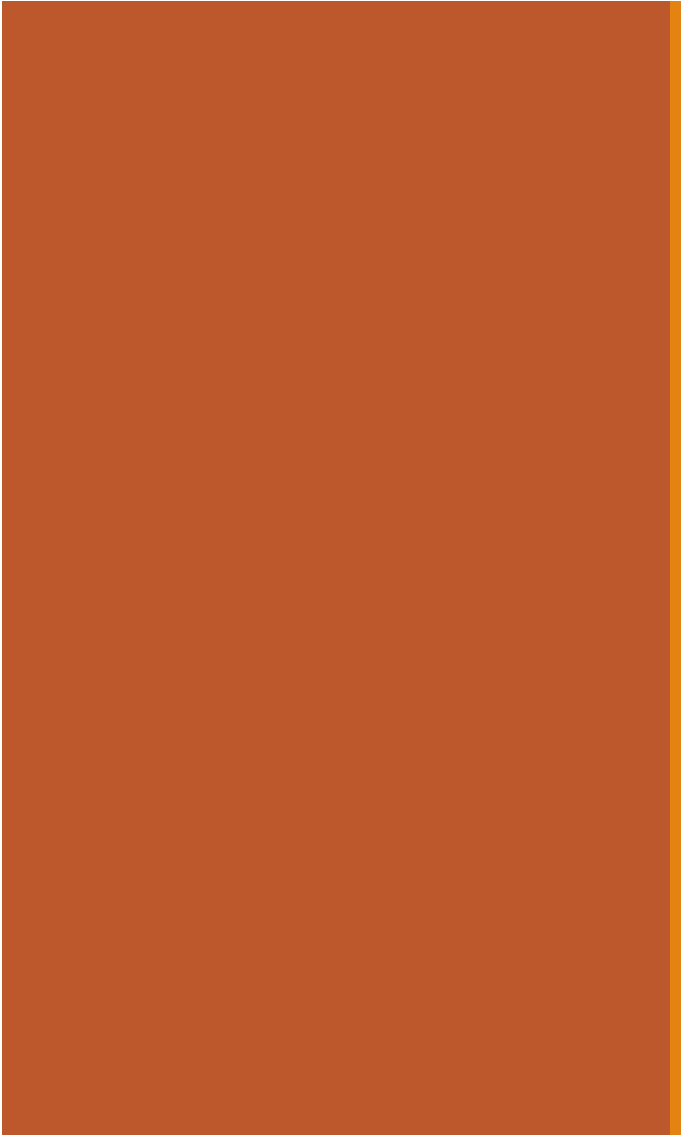
$$P(T = 1|F_{lu} = 0, U = 0) = 0.1$$
$$P(T = 1|F_{lu} = 0, U = 1) = 0.8$$
$$P(T = 1|F_{lu} = 1, U = 0) = 0.9$$
$$P(T = 1|F_{lu} = 1, U = 1) = 1.0$$

What would a CS109 student do?

1. Populate a Bayesian network by asking an infectious diseases expert or by using reasonable assumptions

2. Answer inference questions

Our focus today

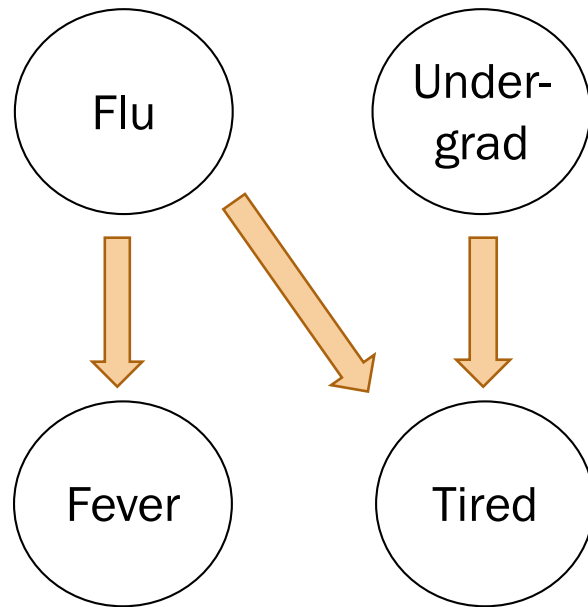
A large, solid orange rectangle occupies the left side of the slide, extending from the top to the bottom and from the left edge to about one-third of the way across the slide.

Inference: Math

Inference via math

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

$$P(T = 1 | F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1 | F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1 | F_{lu} = 1, U = 0) = 0.9$$

$$P(T = 1 | F_{lu} = 1, U = 1) = 1.0$$

1. $P(F_{lu} = 0, U = 1, F_{ev} = 0, T = 1)$

Compute joint probabilities using chain rule.

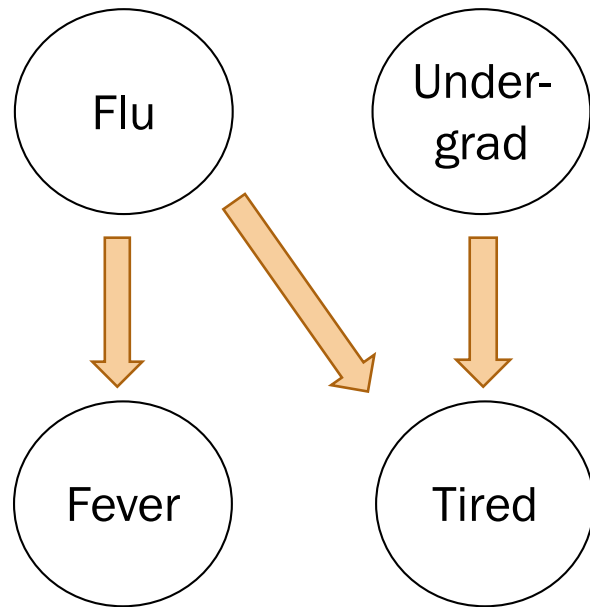
$$\begin{aligned}
 P(F_{lu} = 0) &= 0.9 \\
 P(U = 1 | F_{lu} = 0) &= P(U = 1) = 0.8 \\
 P(F_{ev} = 0 | U = 1, F_{lu} = 0) &= P(F_{ev} = 0 | F_{lu} = 0) = 0.95 \\
 P(T = 1 | F_{ev} = 0, U = 1, F_{lu} = 0) &= P(T = 1 | U = 1, F_{lu} = 0) = 0.8 \\
 \hline
 &= 0.5472
 \end{aligned}$$

all four variable contribute to computation but how they contribute depends on what they are conditioned on.

Inference via math

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1|F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1|F_{lu} = 0) = 0.05$$

$$P(T = 1|F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1|F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1|F_{lu} = 1, U = 0) = 0.9$$

$$P(T = 1|F_{lu} = 1, U = 1) = 1.0$$

2. $P(F_{lu} = 1|F_{ev} = 0, U = 0, T = 1)$?

1. Compute joint probabilities

$$P(F_{lu} = 1, F_{ev} = 0, U = 0, T = 1)$$

$$P(F_{lu} = 0, F_{ev} = 0, U = 0, T = 1)$$

2. Definition of conditional probability

$$\frac{P(F_{lu} = 1, F_{ev} = 0, U = 0, T = 1)}{\sum_x P(F_{lu} = x, F_{ev} = 0, U = 0, T = 1)}$$

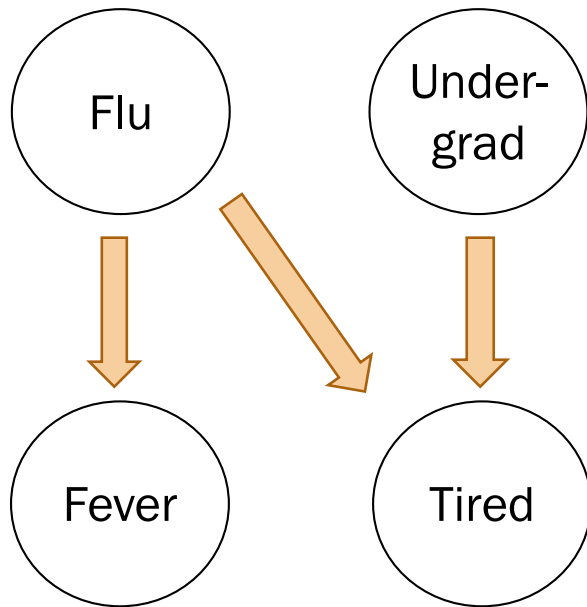
denominator uses LPT over one unknown

$$= 0.095$$

Inference via math

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



3. $P(F_{lu} = 1 | U = 1, T = 1)$?

$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

$$P(T = 1 | F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1 | F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1 | F_{lu} = 1, U = 0) = 0.9$$

$$P(T = 1 | F_{lu} = 1, U = 1) = 1.0$$

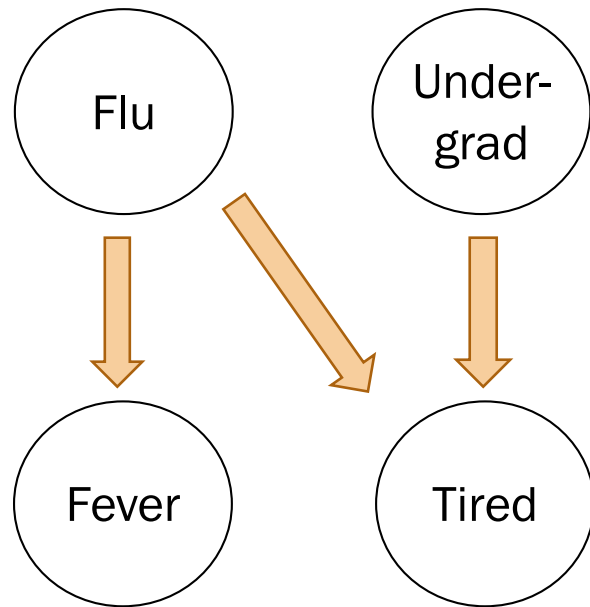
Lisa Yan, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Spring 2024



Inference via math

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1|F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1|F_{lu} = 0) = 0.05$$

$$P(T = 1|F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1|F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1|F_{lu} = 1, U = 0) = 0.9$$

$$P(T = 1|F_{lu} = 1, U = 1) = 1.0$$

$$3. P(F_{lu} = 1|U = 1, T = 1)?$$

1. Compute joint probabilities

there are four values here

$$\left\{ \begin{array}{l} P(F_{lu} = 1, U = 1, F_{ev} = 1, T = 1) \\ \dots \\ P(F_{lu} = 0, U = 1, F_{ev} = 0, T = 1)? \end{array} \right.$$

2. Definition of conditional probability

$$\frac{\sum_y P(F_{lu} = 1, U = 1, F_{ev} = y, T = 1)}{\sum_x \sum_y P(F_{lu} = x, U = 1, F_{ev} = y, T = 1)}$$

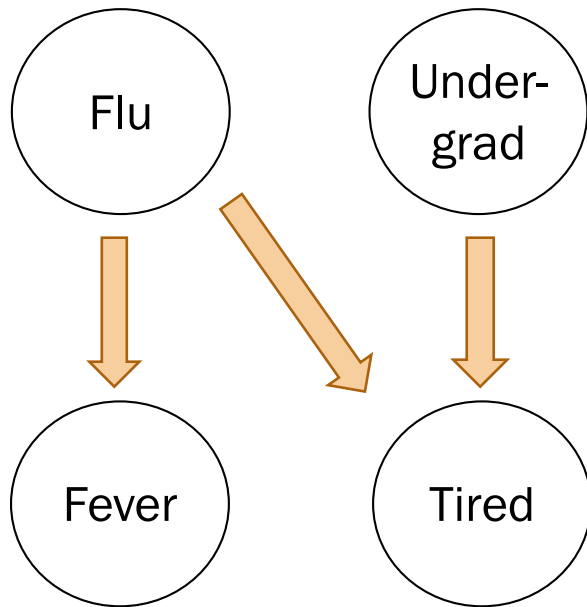
$$= 0.122$$

this one ignores fever, though its computation doesn't

Inference via math

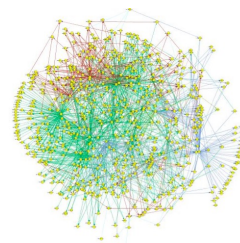
$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$
$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

$$P(T = 1 | F_{lu} = 0, U = 0) = 0.1$$
$$P(T = 1 | F_{lu} = 0, U = 1) = 0.8$$
$$P(T = 1 | F_{lu} = 1, U = 0) = 0.9$$
$$P(T = 1 | F_{lu} = 1, U = 1) = 1.0$$

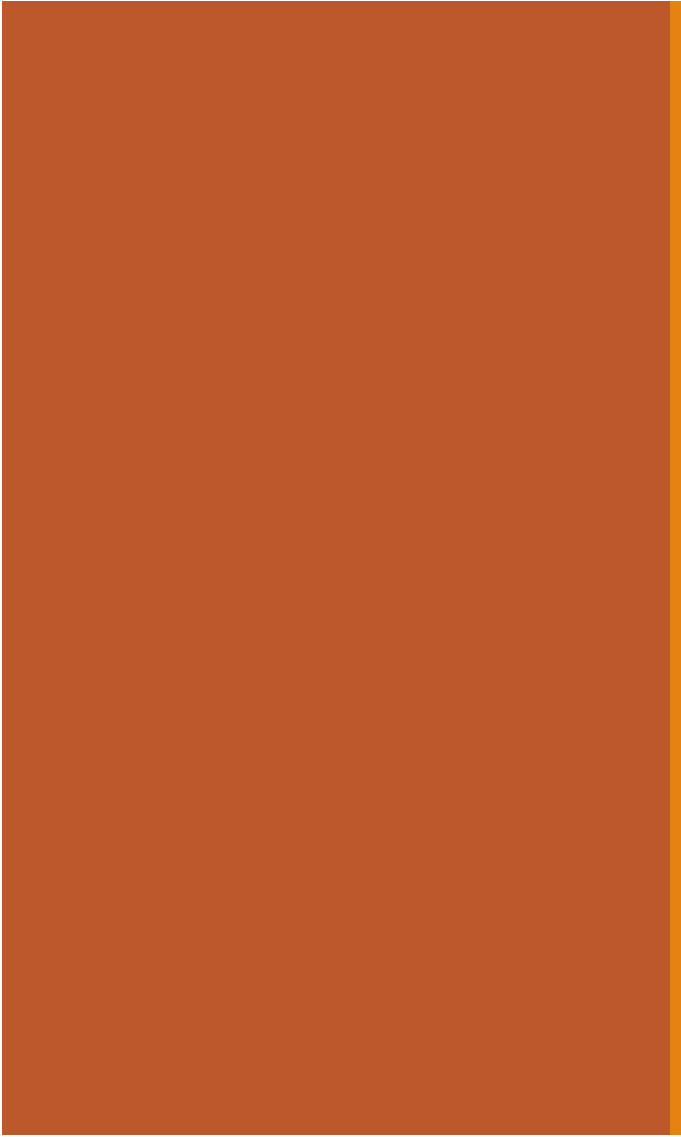


Solving inference questions precisely is possible, but sometimes tedious.

often :)

Can we use sampling to do approximate inference?

yasss



Rejection Sampling

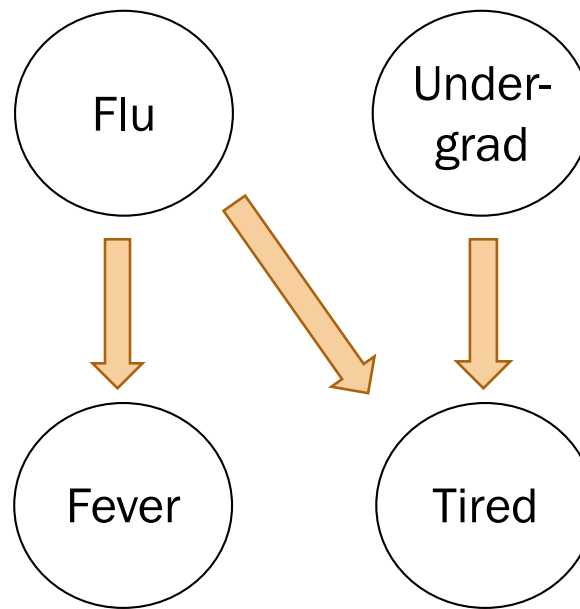
Rejection sampling algorithm

Step 0:

Require a fully specified
Bayesian Network

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

$$P(T = 1 | F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1 | F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1 | F_{lu} = 1, U = 0) = 0.9$$

$$P(T = 1 | F_{lu} = 1, U = 1) = 1.0$$

Rejection sampling algorithm

Inference
question:

What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation = ...  
    # number of samples with (U = 1, T = 1)  
    samples_event =  
    # number of samples with (Flu = 1, U = 1, T = 1)  
    return len(samples_event) / len(samples_observation)
```

[flu, und, fev, tir]

```
Sampling...  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 0, 0, 0]  
[0, 1, 0, 1]  
[0, 1, 1, 1]  
[0, 1, 0, 0]  
[1, 1, 1, 1]  
[0, 0, 1, 1]  
...  
[0, 1, 0, 1]  
Finished sampling
```

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation = ...  
        # number of samples with  $(U = 1, T = 1)$   
    samples_event =  
        # number of samples with  $(F_{lu} = 1, U = 1, T = 1)$   
    return len(samples_event) / len(samples_observation)
```

$$\text{Probability} \approx \frac{\text{\# samples with } (F_{lu} = 1, U = 1, T = 1)}{\text{\# samples with } (U = 1, T = 1)}$$

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation = ...  
    # number of samples with (U = 1, T = 1)  
    samples_event = ...  
    # number of samples with (Flu = 1, U = 1, T = 1)  
    return len(samples_event) / len(samples_observation)
```

[flu, und, fev, tir]

```
Sampling...  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 0, 0, 0]  
[0, 1, 0, 1]  
[0, 1, 1, 1]  
[0, 1, 0, 0]  
[1, 1, 1, 1]  
[0, 0, 1, 1]  
...  
[0, 1, 0, 1]  
Finished sampling
```

Rejection sampling algorithm

```
N_SAMPLES = 100000
# Method: Sample a ton
# -----
# create N_SAMPLES with likelihood proportional
# to the joint distribution
def sample_a_ton():
    samples = []
    for i in range(N_SAMPLES):
        sample = make_sample() # a particle
        samples.append(sample)
    return samples
```

How do we make a sample
 $(F_{lu} = a, U = b, F_{ev} = c, T = d)$
according to the
joint probability?

Create a sample using the Bayesian Network!

Rejection sampling algorithm

```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

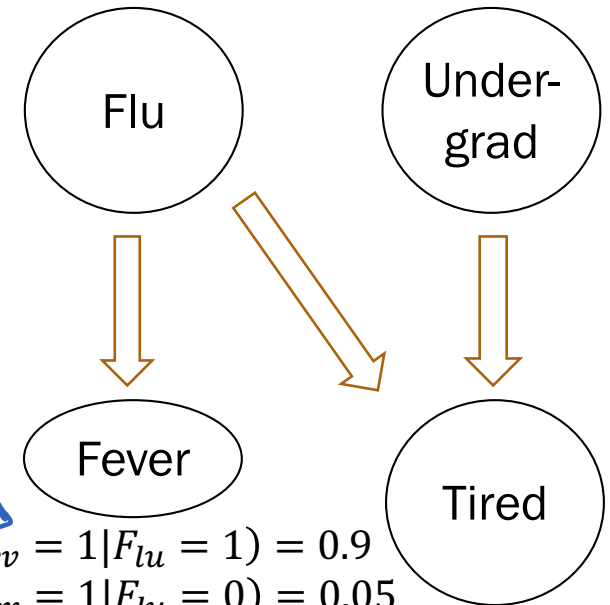
    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else:         fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    #
    # TODO: fill in
    #
    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tired]
```

suggests about 10% of samples should set flu=1

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

this says about 90% of sample where flu=1 should have fever=1 as well.

$$P(T = 1 | F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1 | F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1 | F_{lu} = 1, U = 0) = 0.9$$

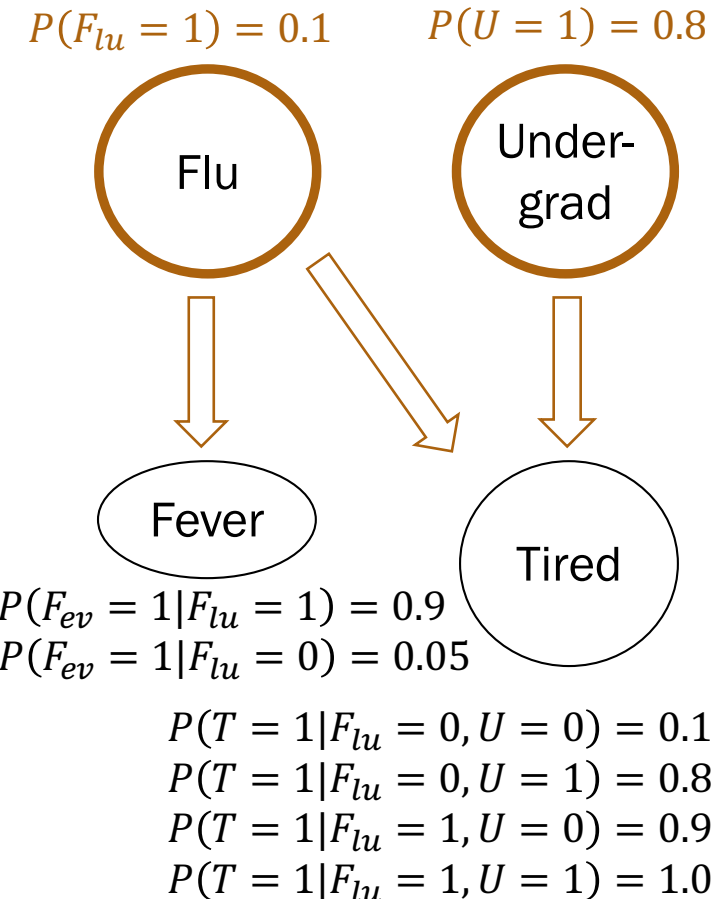
$$P(T = 1 | F_{lu} = 1, U = 1) = 1.0$$

Rejection sampling algorithm

```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    #
    # TODO: fill in
    #
    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tired]
```

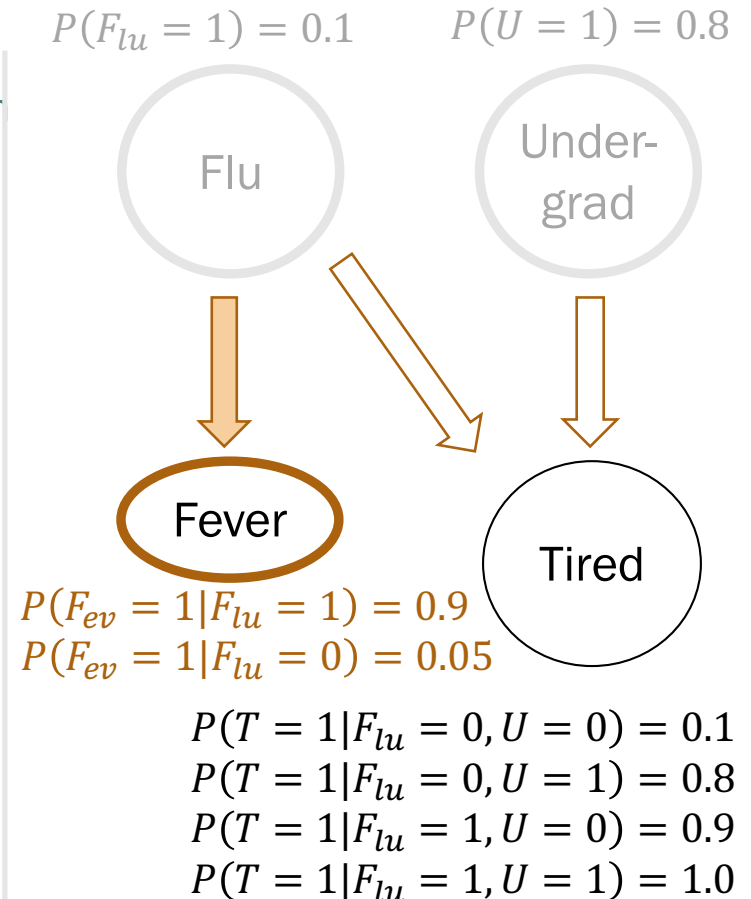


Rejection sampling algorithm

```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    #
    # TODO: fill in
    #
    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tired]
```



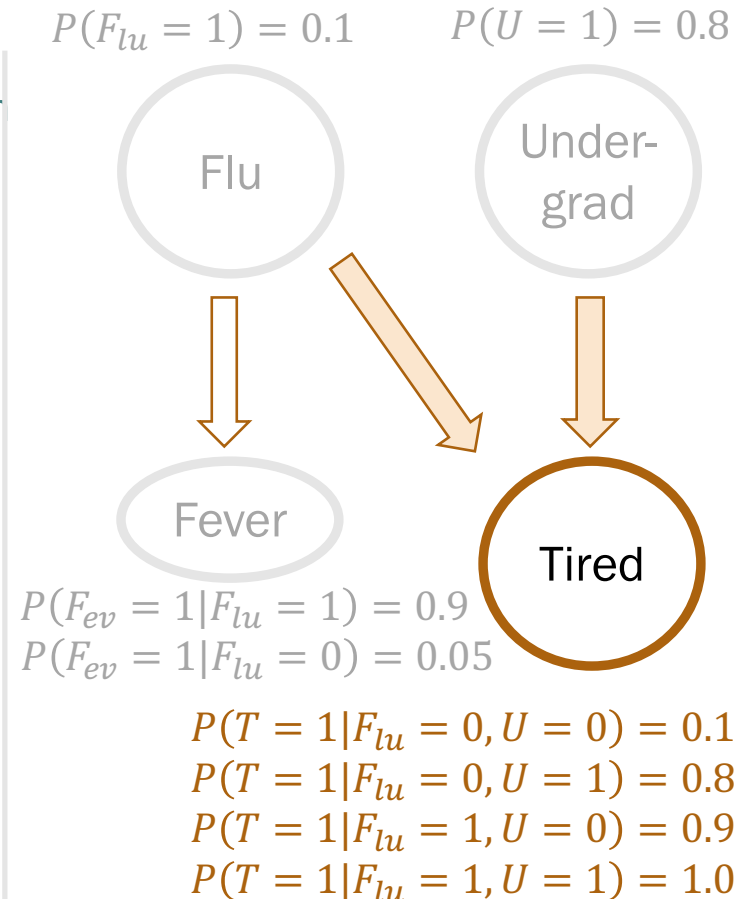
Rejection sampling algorithm

```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    #
    # TODO: fill in
    #
    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tired]
```

Lisa Yan, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Spring 2024



Rejection sampling algorithm

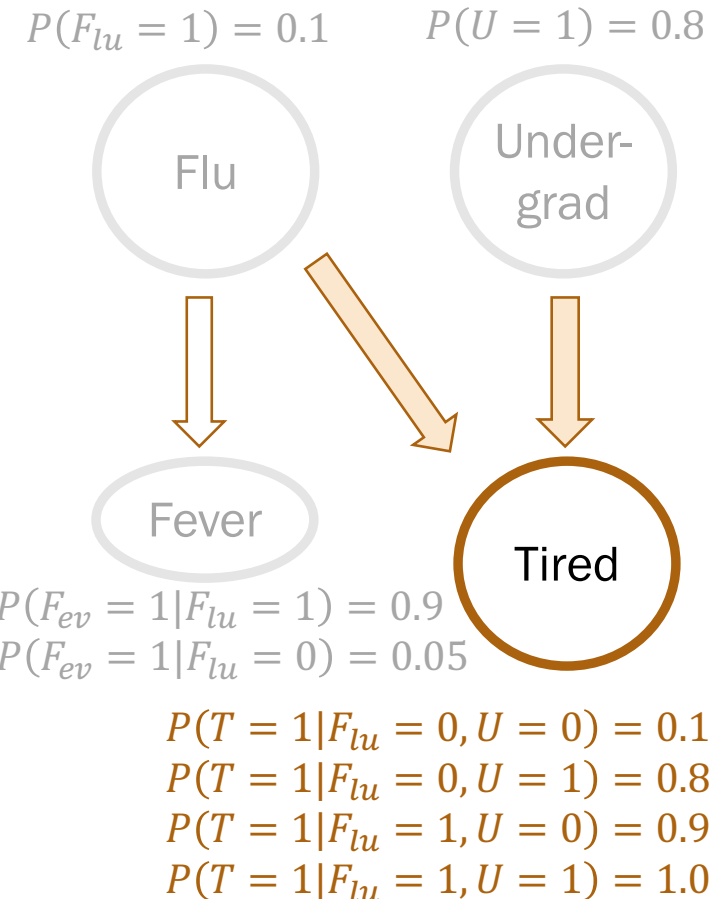
```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    if flu == 0 and und == 0: tir = bernoulli(0.1)
    elif flu == 0 and und == 1: tir = bernoulli(0.8)
    elif flu == 1 and und == 0: tir = bernoulli(0.9)
    else: tir = bernoulli(1.0)

    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tir]
```

Lisa Yan, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Spring 2024



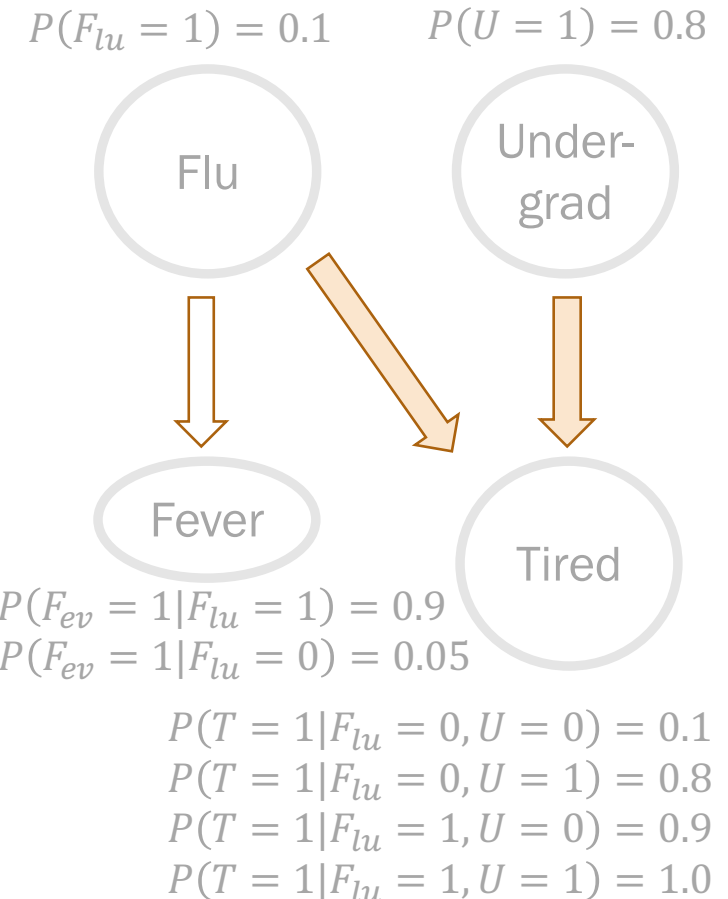
Rejection sampling algorithm

```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    if flu == 0 and und == 0: tir = bernoulli(0.1)
    elif flu == 0 and und == 1: tir = bernoulli(0.8)
    elif flu == 1 and und == 0: tir = bernoulli(0.9)
    else: tir = bernoulli(1.0)

    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tir]
```



Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation = ...  
    # number of samples with (U = 1, T = 1)  
    samples_event = ...  
    # number of samples with (Flu = 1, U = 1, T = 1)  
    return len(samples_event) / len(samples_observation)
```

[flu, und, fev, tir]

```
Sampling...  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 0, 0, 0]  
[0, 1, 0, 1]  
[0, 1, 1, 1]  
[0, 1, 0, 0]  
[1, 1, 1, 1]  
[0, 0, 1, 1]  
...  
[0, 1, 0, 1]  
Finished sampling
```

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation = ...  
        # number of samples with  $(U = 1, T = 1)$   
    samples_event =  
        # number of samples with  $(F_{lu} = 1, U = 1, T = 1)$   
    return len(samples_event)/len(samples_observation)
```

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation =  
        reject_inconsistent(samples, observation)  
    samples_event =  
        # number of samples with  $(F_{lu} = 1, U = 1, T = 1)$   
    return len(samples_event) / len(samples_observation)
```

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation =  
        reject_inconsistent(samples, observation)  
    samples_event =  
        # number of samples with  $(F_{lu} = 1, U = 1, T = 1)$   
    return len(samples_event) / len(samples_observation)
```

Keep only samples that are consistent
with the observation $(U = 1, T = 1)$.

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):
    samples = sample_a_ton()
    samples_observation =
        reject_inconsistent(samples, observation)
    samples
    # Method: reject_inconsistent
    # -----
    # Rejects all samples that do not align with the outcome.
    # Returns a list of consistent samples.
    return
    def reject_inconsistent(samples, outcome):
        consistent_samples = []
        for sample in samples:
            if check_consistent(sample, outcome):
                consistent_samples.append(sample)
        return consistent_samples
```

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation =  
        reject_inconsistent(samples, observation)  
    samples_event =  
        reject_inconsistent(samples_observation, event)  
    return len(samples_event)/len(samples_observation)
```

Conditional event = samples with $(F_{lu} = 1, U = 1, T = 1)$.

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):
    samples = sample_a_ton()
    samples_observation =
        reject_inconsistent(samples, observation)
    samples_event =
        reject_inconsistent(samples_observation, event)
    return 1
```

Condition: $(F_{lu} = x, U = 1, F_{ev} = y, T = 1) \rightarrow (F_{lu} = 1) = 1$.

```
def reject_inconsistent(samples, outcome):
    return consistent_samples
```

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation =  
        reject_inconsistent(samples, observation)  
    samples_event =  
        reject_inconsistent(samples_observation, event)  
    return len(samples_event)/len(samples_observation)
```

$$\text{Probability} \approx \frac{\# \text{ samples with } (F_{lu} = 1, U = 1, T = 1)}{\# \text{ samples with } (U = 1, T = 1)}$$

Rejection sampling

If you can sample enough from the joint distribution, you can answer any probability inference question.

With enough samples, you can easily compute:

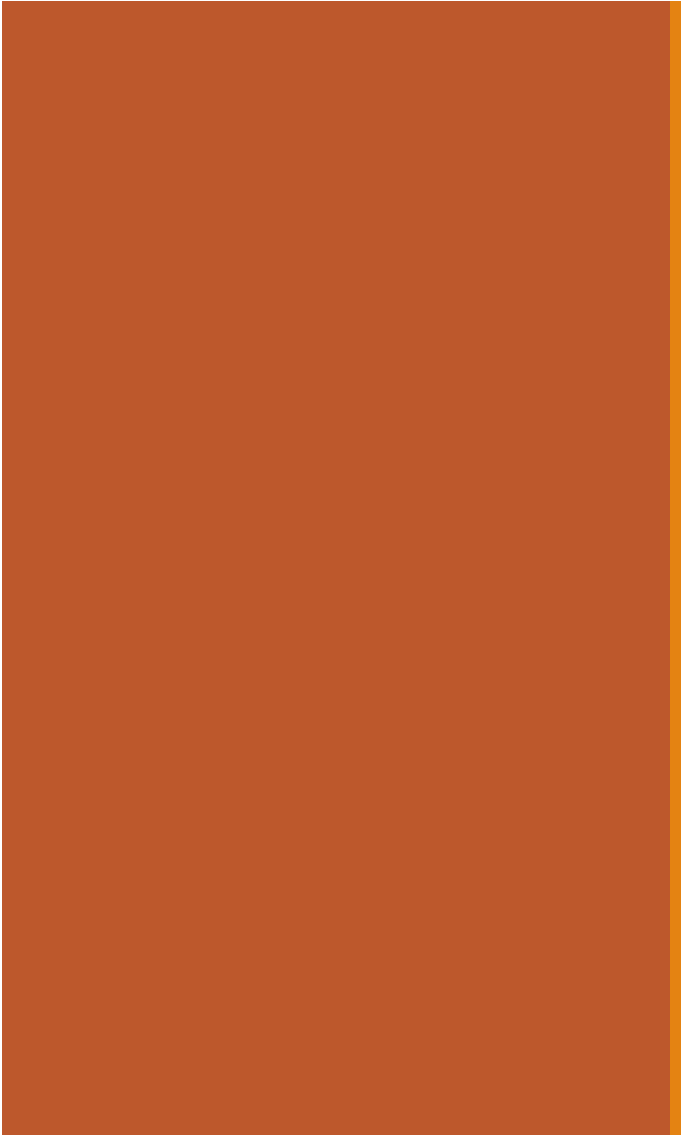
- Probability estimates
- Conditional probability estimates
- Expectation estimates

Why? Because your samples represent the joint distribution incredibly well!

[flu, und, fev, tir]

```
Sampling...  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 0, 0, 0]  
[0, 1, 0, 1]  
[0, 1, 1, 1]  
[0, 1, 0, 0]  
[1, 1, 1, 1]  
[0, 0, 1, 1]  
...  
[0, 1, 0, 1]  
Finished sampling
```

$$P(\text{has flu} \mid \text{undergrad and is tired}) = 0.122$$

A large, solid orange rectangle occupies the left side of the slide, extending from the top to the bottom and from the left edge to approximately one-third of the way across the slide.

Breaking News

Before You Go! CS109 Challenge



Do something cool and creative
with probability!

Grand Prize:

All exams replaced with 100%

All Serious Entries:

Extra credit (between %0.5 and
2% added to overall average)

Optional Proposal: Sun. 05/12, 11:59pm

Due: Sun. 06/02, 11:59pm

<https://cs109.stanford.edu/handouts/challenge.html>